

# Supplemental Material for

## Mutation-process priors calibrate protein language models for retrospective genotype–phenotype evolutionary ranking

### S1 Codon-transition model

#### S1.1 Amino-acid substitution probability induced by nucleotide dynamics

We model a viral genome as a continuous-time Markov process (CTMP) on  $\mathcal{B}^N$ , where  $\mathcal{B} = \{\text{A, C, G, U}\}$  is the RNA alphabet. The model assumes that sites evolve independently and that the transition rate at a site depends only on the current and mutated base. For a single base, let  $Q = [q_{uv}]_{u,v \in \mathcal{B}}$  be the infinitesimal generator, with  $q_{uu} = -\sum_{v \neq u} q_{uv}$ . The transition probability over time  $T$  is

$$\mathbb{P}(b_{t+T} = v \mid b_t = u) = [\exp(TQ)]_{uv}. \quad (1)$$

For a codon  $\mathbf{c} = (c_1, c_2, c_3)$ , independence of the three nucleotide positions gives

$$\mathbb{P}(\mathbf{c}_{t+T} = \tilde{\mathbf{c}} \mid \mathbf{c}_t = \mathbf{c}) = \prod_{k=1}^3 [\exp(TQ)]_{c_k \tilde{c}_k}. \quad (2)$$

Let  $g(\mathbf{c})$  denote the amino acid encoded by codon  $\mathbf{c}$ . The codon-induced probability of producing amino acid  $\tilde{a}$  from wild-type codon  $\mathbf{c}$  is therefore

$$p_T(\tilde{a} \mid \mathbf{c}) = \sum_{\tilde{\mathbf{c}}: g(\tilde{\mathbf{c}}) = \tilde{a}} \prod_{k=1}^3 [\exp(TQ)]_{c_k \tilde{c}_k}. \quad (3)$$

This is the CLIMB process prior used in the main text.

#### S1.2 Estimating the nucleotide transition matrix

For an ergodic CTMP, the long-run transition flux  $f_{uv}$  from base  $u$  to base  $v$  satisfies  $f_{uv} = \pi_u q_{uv}$ , where  $\pi_u$  is the stationary frequency [1]. We estimate

$$\hat{\pi}_u = \frac{n_u}{\sum_{u' \in \mathcal{B}} n_{u'}}, \quad \hat{f}_{uv} = \frac{n_{uv}}{\sum_{u' \neq v'} n_{u'v'}}, \quad \hat{q}_{uv} = \frac{\hat{f}_{uv}}{\hat{\pi}_u} \quad (u \neq v). \quad (4)$$

Rates are normalized so that  $\sum_{u \neq v} \hat{f}_{uv} = 1$ . Using the same stationary base frequencies as the Hie et al. sequence set [2],

$$\hat{\pi} = \begin{bmatrix} \text{A} & \text{C} & \text{G} & \text{U} \\ 0.2994 & 0.1837 & 0.1961 & 0.3208 \end{bmatrix}. \quad (5)$$

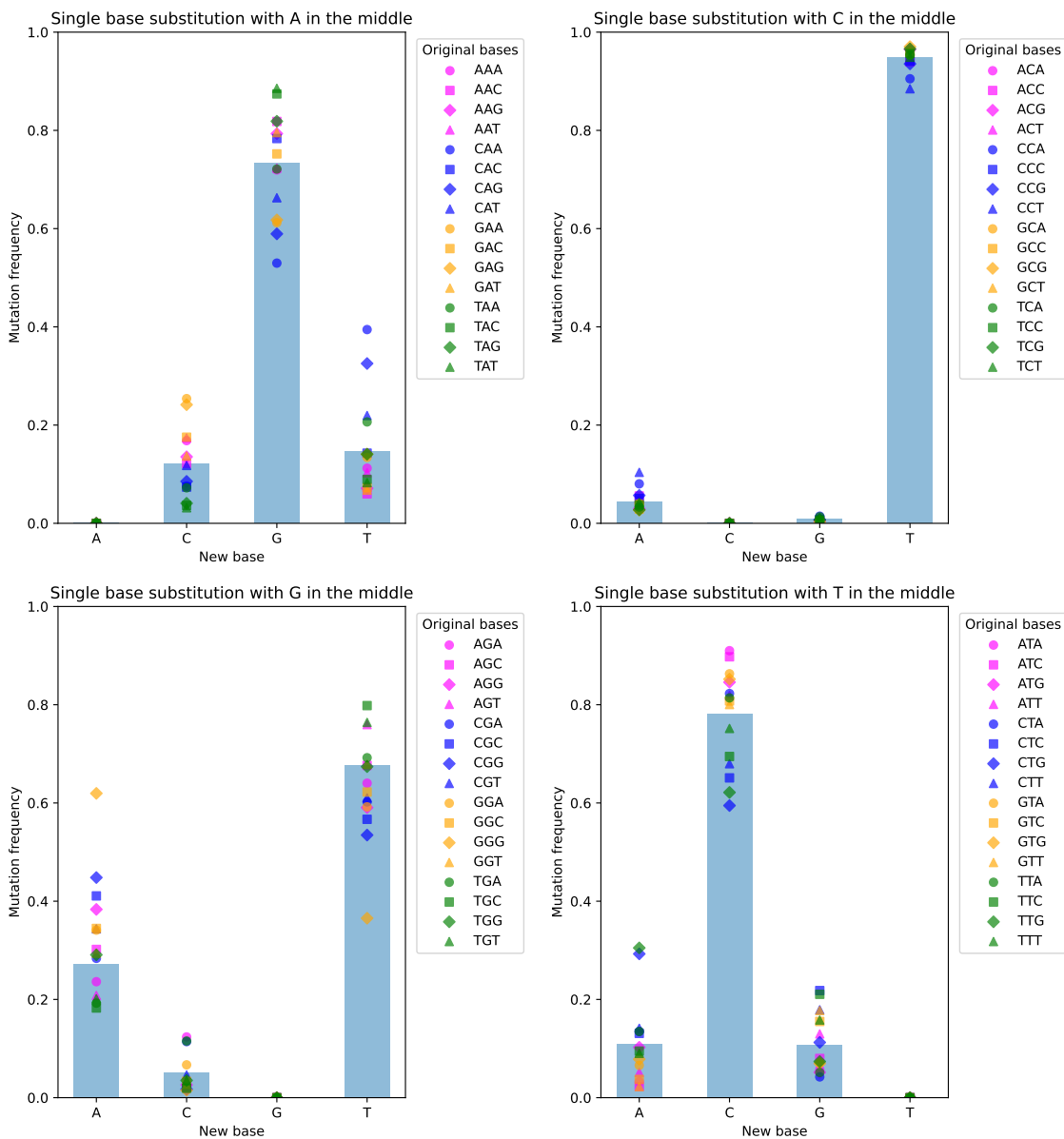
Combining these frequencies with substitution counts inferred from SARS-CoV-2 mutational data [3] gives

$$\hat{Q} = \begin{bmatrix} -0.3995 & 0.0480 & 0.2932 & 0.0583 \\ 0.1183 & -2.6735 & 0.0215 & 2.5338 \\ 0.3742 & 0.0693 & -1.3716 & 0.9281 \\ 0.0414 & 0.2934 & 0.0405 & -0.3753 \end{bmatrix}, \quad (6)$$

with row and column order (A, C, G, U).

### S1.3 Context-independence check

The main model uses a context-averaged nucleotide process. To test whether this is a reasonable first-order approximation, we compared unconditional base-change probabilities with probabilities conditioned on the immediate 5' and 3' nucleotide contexts. The high C→U rate is stable across contexts, whereas A→G and G→A show stronger context variation. Thus, the context-averaged model captures the dominant central tendency but should be replaced by a context-dependent generator in future prospective versions.



**Figure S1: Unconditional and context-conditioned substitution probabilities.** Bars show unconditional probabilities for each central-base substitution. Points show probabilities conditioned on the 16 possible trinucleotide contexts.

## S2 CAC scoring and optimization

### S2.1 Score definition

The main text defines CAC as a weighted mixture of standardized terms,

$$S_{\text{CAC}} = \alpha z(s_s) + \beta z(\log s_g) + \gamma z(\log p_T), \quad \alpha, \beta, \gamma \geq 0, \quad \alpha + \beta + \gamma = 1. \quad (7)$$

Here  $s_s$  is semantic change,  $s_g$  is grammaticality, and  $p_T$  is the codon-transition prior in Eq. (3). The standardization  $z(\cdot)$  is performed over all nonsynonymous single-nucleotide substitutions of the reference Spike sequence.

### S2.2 Validation sets

Weights were optimized by minimizing the escape mean rank (EMR) on a specified validation set. For a set  $\mathcal{V}$  of known mutations,

$$\text{EMR}(\mathcal{V}; \alpha, \beta, \gamma) = \frac{1}{|\mathcal{V}|} \sum_{m \in \mathcal{V}} \text{rank}(m \mid \alpha, \beta, \gamma). \quad (8)$$

The current results are retrospective benchmarks. A strict prospective deployment should use time-split or leave-one-variant-out tuning to separate weight selection from final testing.

**Table S1: Validation sets used for retrospective optimization and evaluation.**

| Name     | Point mutations                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DMS      | K417E, K444Q, V445A, N450D, Y453F, L455F, E484K, G485D, F486V, F490L, F490S, Q493K, H655Y, R682Q, R685S, V687G, G769E, Q779K, V1128A                                                                                                                                                                                                                                                                                            |
| Alpha    | N501Y, A570D, D614G, P681H, T716I, S982A, D1118H                                                                                                                                                                                                                                                                                                                                                                                |
| Beta     | L18F, D80A, D215G, K417N, E484K, N501Y, D614G, A701V                                                                                                                                                                                                                                                                                                                                                                            |
| Delta    | T19R, G142D, L452R, T478K, D614G, P681R, D950N                                                                                                                                                                                                                                                                                                                                                                                  |
| Omicron  | A67V, T95I, G142D, L212I, G339D, S371L, S373P, S375F, K417N, N440K, G446S, S477N, T478K, E484A, Q493R, G496S, Q498R, N501Y, Y505H, T547K, D614G, H655Y, N679K, P681H, N764K, D796Y, N856K, Q954H, N969K, L981F                                                                                                                                                                                                                  |
| Combined | A570D, A67V, A701V, D1118H, D215G, D614G, D796Y, D80A, D950N, E484A, E484K, F486V, F490L, F490S, G142D, G339D, G446S, G485D, G496S, G769E, H655Y, K417E, K417N, K444Q, L18F, L212I, L452R, L455F, L981F, N440K, N450D, N501Y, N679K, N764K, N856K, N969K, P681H, P681R, Q493K, Q493R, Q498R, Q779K, Q954H, R682Q, R685S, S371L, S373P, S375F, S477N, S982A, T19R, T478K, T547K, T716I, T95I, V1128A, V445A, V687G, Y453F, Y505H |

### S2.3 Simplex search for optimal weights

The rank transform makes EMR piecewise constant and non-differentiable. We therefore used recursive subdivision of the simplex  $\Delta^2 = \{(\alpha, \beta, \gamma) : \alpha, \beta, \gamma \geq 0, \alpha + \beta + \gamma = 1\}$ . At each iteration the current triangle is divided into four subtriangles (Fig. S2). For each subtriangle  $T$  with vertices  $v_1, v_2, v_3$  and centroid  $c$ , we evaluate

$$S(T) = \frac{1}{12} [\text{EMR}(v_1) + \text{EMR}(v_2) + \text{EMR}(v_3) + 9 \text{EMR}(c)], \quad (9)$$

and continue with the subtriangle having the smallest  $S(T)$ .

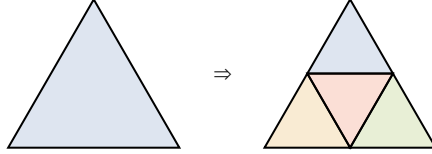


Figure S2: Recursive subdivision of the coefficient simplex.

### S3 Label-free calibration by phylogenetic fitness

The main-text weights can be fixed without escape labels. We reparameterize the score by freezing the codon-accessibility term at unit coefficient (an *offset*) and learning only the two protein-signal weights,

$$S = b_0 z(s_s) + b_1 z(\log s_g) + \kappa z(\log p_T), \quad \kappa = 1, \quad (10)$$

and determine  $(b_0, b_1)$  by logistic regression of the standardized protein terms against a binarized *phylogenetic fitness* target, with the codon term entering as a fixed offset. The target is the Bloom–Neher per-mutation fitness  $\Delta f = \ln[(n_{\text{actual}} + P)/(n_{\text{expected}} + P)]$ , where expected counts follow the neutral mutation spectrum at four-fold-degenerate third-codon-position sites on the SARS-CoV-2 tree;  $\Delta f$  is estimated from natural sequences independently of the DMS escape benchmark. No escape label enters the fit.

Table S2 compares the label-free weights and held-out DMS escape AUROC against escape-supervised weights (obtained by fitting the same form to DMS labels) across backbones. For the domain-adapted ESM-2<sub>cov</sub> the label-free weights (0.285, 0.942) recover the supervised weights (0.268, 0.955) and match its DMS AUROC; for general-purpose ESM-2, whose raw protein features carry little escape signal, the fit correctly drives the protein weights toward zero so that the score falls back on the mutation-process term without harm.

**Table S2: Label-free vs escape-supervised calibration across backbones.** Weights  $(b_0, b_1)$  on semantic change and grammaticality (codon prior frozen,  $\kappa = 1$ ); DMS escape AUROC. “bloom-fit” is fit to phylogenetic fitness (no escape labels); “supervised” is fit to DMS labels (reference).

| Backbone             | Method           | $b_0$ (sem.) | $b_1$ (gram.) | DMS AUROC    |
|----------------------|------------------|--------------|---------------|--------------|
| base ESM-2           | equal (1, 1)     | 1.000        | 1.000         | 0.786        |
|                      | CLIB-only        | 0.000        | 0.000         | 0.883        |
|                      | bloom-fit        | −0.102       | 0.449         | 0.863        |
|                      | supervised (DMS) | 0.094        | 0.313         | 0.880        |
| ESM-2 <sub>cov</sub> | equal (1, 1)     | 1.000        | 1.000         | 0.884        |
|                      | CLIB-only        | 0.000        | 0.000         | 0.883        |
|                      | <b>bloom-fit</b> | <b>0.285</b> | <b>0.942</b>  | <b>0.899</b> |
|                      | supervised (DMS) | 0.268        | 0.955         | 0.898        |

Three independent evolutionary signals yield concordant weights on ESM-2<sub>cov</sub> (Table S3): all-time substitution frequency, phylogenetic fitness, and an early-period ( $\leq 2021$ ) temporal split. All favor grammaticality over semantic change and reproduce the supervised operating point, indicating that the calibration is a property of the mutation process rather than of any single proxy.

**Table S3: Cross-signal convergence of label-free weights (ESM-2<sub>cov</sub>).**

| Fitting signal         | $b_0$ (sem.) | $b_1$ (gram.) | DMS AUROC |
|------------------------|--------------|---------------|-----------|
| Frequency (all-time)   | 0.191        | 1.543         | 0.883     |
| Phylogenetic fitness   | 0.285        | 0.942         | 0.899     |
| Temporal (early split) | 0.194        | 1.681         | 0.881     |

## S4 Prospective and leave-one-variant-out validation

A calibration without escape labels can be made strictly prospective. We fit  $(b_0, b_1)$  on substitution frequencies observed before a date cutoff and evaluate on substitutions that rose in frequency afterward. Because the PLM terms are frequency-independent and the future is unavailable at fit time, this is a genuine held-out forecast. Table S4 reports the emergence-forecast AUROC at three cutoffs, with paired-bootstrap 95% confidence intervals for the improvement over an equal-weight baseline. The label-free score forecasts emergent substitutions up to AUROC 0.90, significantly above equal weight at the two earlier cutoffs and comparable to the escape-supervised reference (0.894), which it never sees.

**Table S4: Prospective emergence forecasting.** Weights fit on pre-cutoff frequency; evaluated on post-cutoff emergent substitutions.  $\Delta$ AUROC is label-free minus equal-weight, with paired-bootstrap 95% CI.

| Cutoff  | $n_{\text{emg}}$ | equal | CLIB-only | label-free | $\Delta$ AUROC (95% CI) |
|---------|------------------|-------|-----------|------------|-------------------------|
| 2021-06 | 25               | 0.940 | 0.885     | 0.976      | +0.036 [0.025, 0.046]   |
| 2021-12 | 15               | 0.860 | 0.836     | 0.899      | +0.038 [0.017, 0.061]   |
| 2022-06 | 34               | 0.807 | 0.838     | 0.824      | +0.018 [−0.003, 0.038]  |

To confirm that the improvement is not memorization of the evaluated annotations, we recalibrated with leave-one-variant-out: for each lineage, its defining substitutions were removed from the fitting set before weights were determined and the lineage was then scored. The gain over equal weight persisted on all tested lineages (each 95% bootstrap CI on the AUROC improvement excluded zero for lineages with  $\geq 7$  positives), so the calibration generalizes to substitutions it never used.

## S5 Model backbones and computational environment

We evaluated four PLM backbones: the viral Hie-BiLSTM model [2] and ESM-2-150M, ESM-2-650M, and ESM-2-3B [4]. For Hie-BiLSTM, we used precomputed semantic and grammaticality scores from the original authors. ESM-2 scores were generated on a server with a single NVIDIA H100 80GB GPU, two AMD EPYC 9124 16-core CPUs, and 768GB RAM, running Ubuntu 24.04.2 LTS. The Python environment used Python 3.10.17, PyTorch 2.7.0, Transformers 4.51.3, and CUDA 12.8.

## S6 Additional performance results

## S7 Additional landscape analysis

The PRL manuscript emphasizes the model-process mismatch and the horizon-dependent crossover. The domain-level Spike landscape analysis from the long-form manuscript is retained here as diagnostic information. These summaries should be interpreted as model-behavior diagnostics rather than independent biological claims.

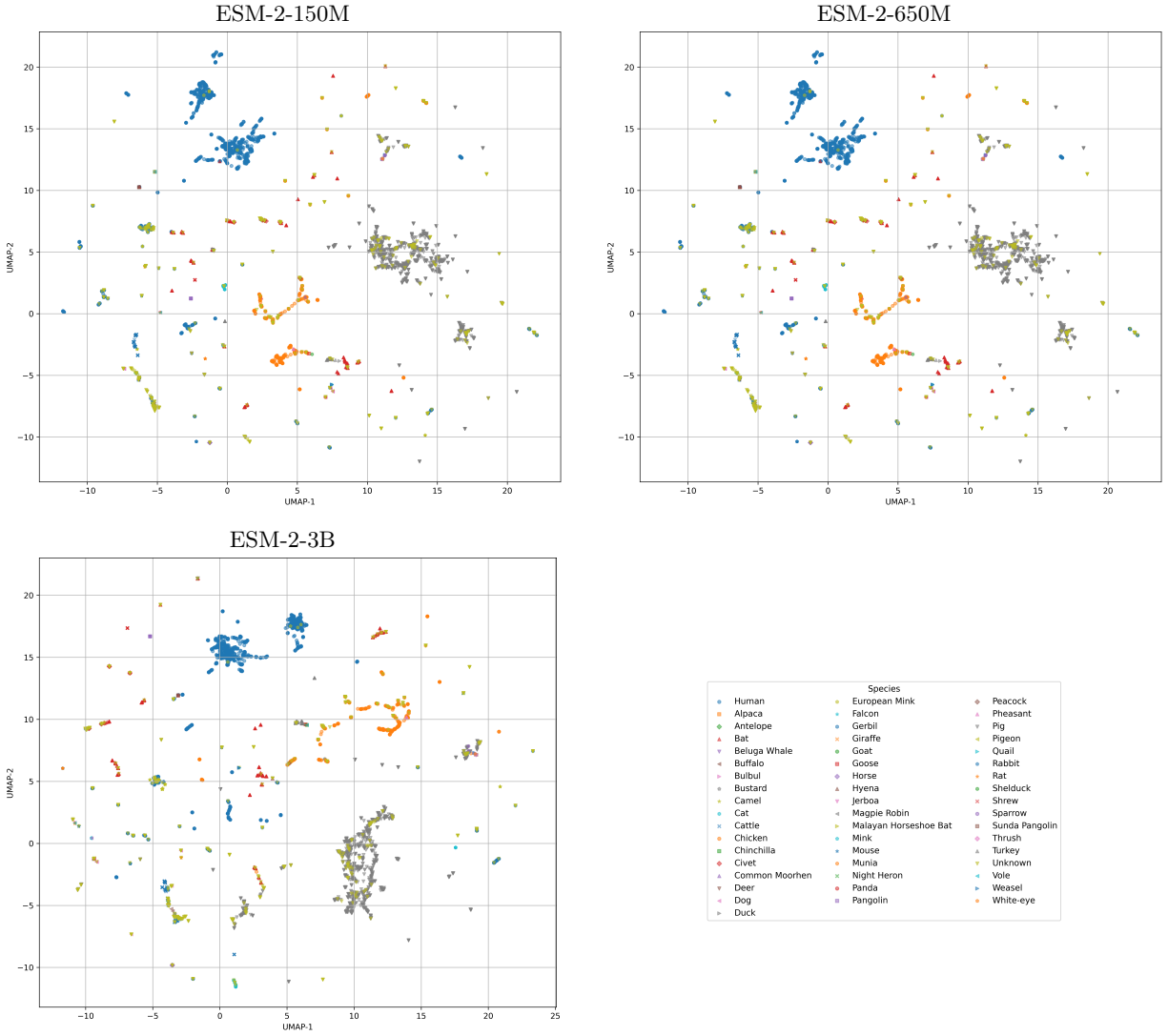
## S8 Availability and reproducibility

All source code developed for this study is available at <https://github.com/sos440/climb> and archived on Zenodo at <https://doi.org/10.5281/zenodo.15744443>. The same Zenodo archive contains the generated CAC scores, CSCS scores, model component scores, and data needed to reproduce the figures and tables.

The core reproduction workflow is:

1. Clone the repository and install dependencies.

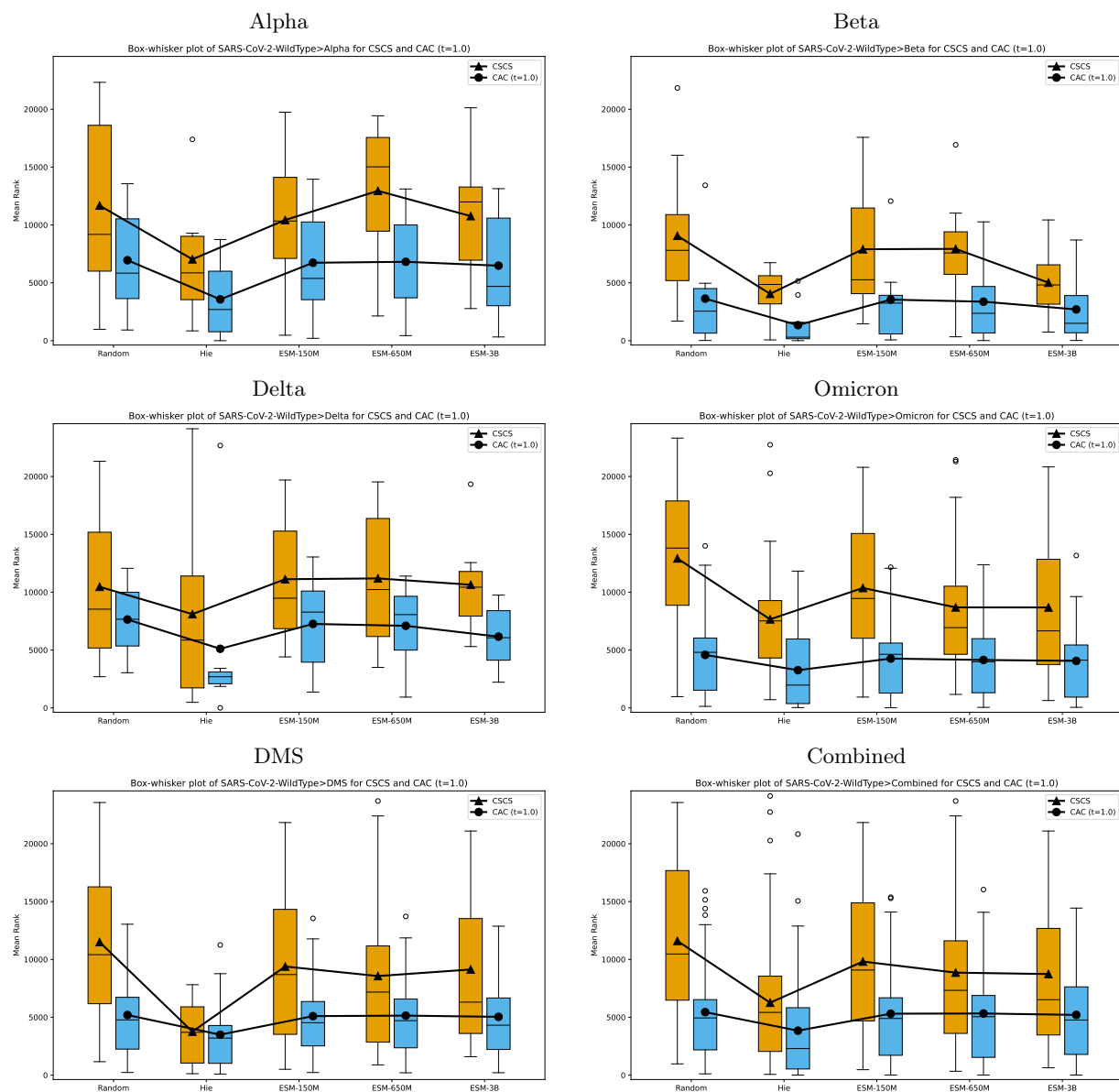
```
$ git clone https://github.com/sos440/climb
$ cd climb
```



**Figure S3: UMAP projections of ESM-2 Spike embeddings.** Points represent unique SARS-CoV-2 Spike sequences colored by host species.

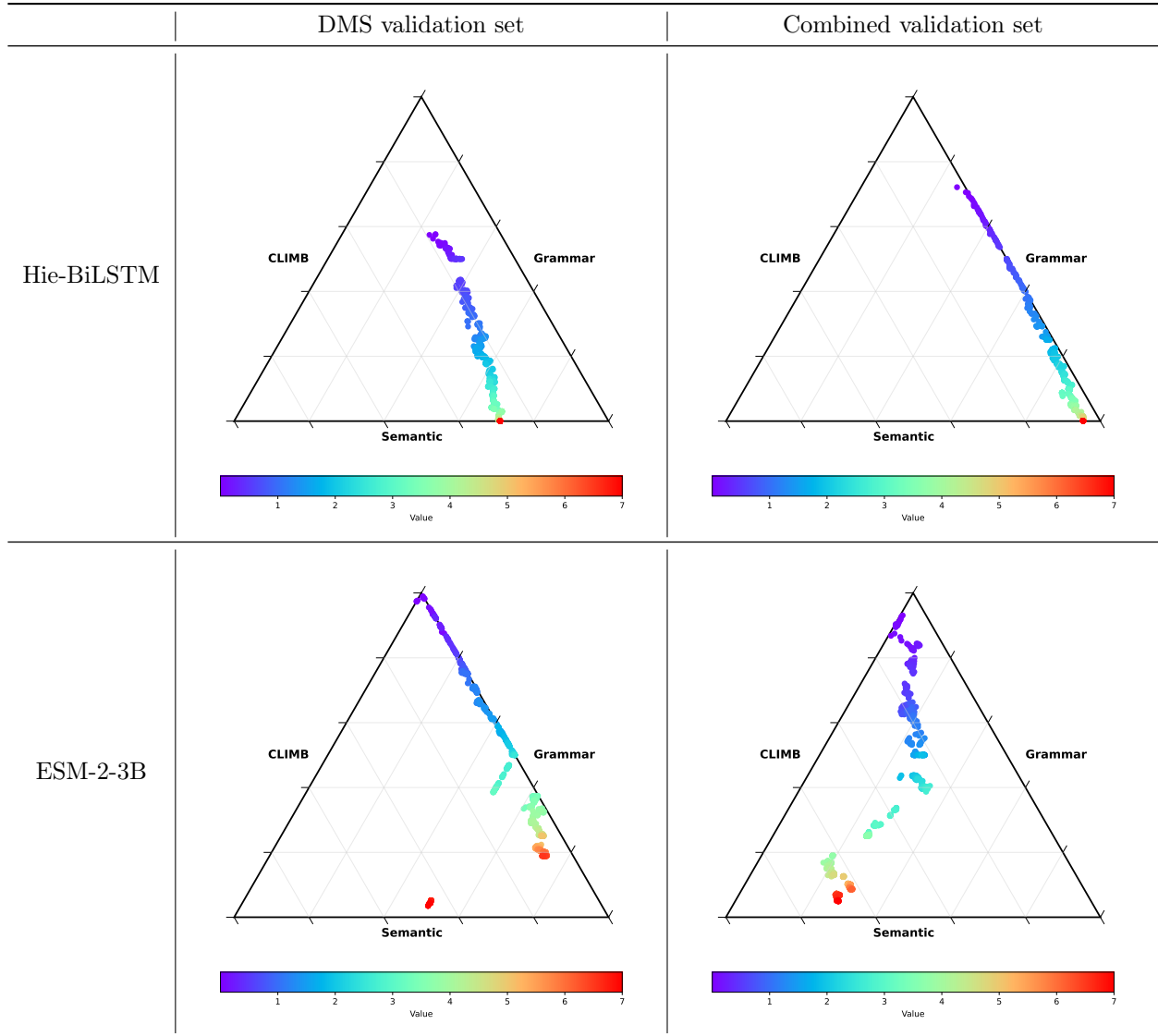
**Table S5: Comprehensive mean-rank comparison of CSCS and CAC.**

| Validation set | Model backbone | Escape rank (mean $\pm$ s.d.) |                       | Wilcoxon $p$ -value  |
|----------------|----------------|-------------------------------|-----------------------|----------------------|
|                |                | CSCS                          | CAC                   |                      |
| DMS            | Hie-BiLSTM     | 3757.16 $\pm$ 2573.64         | 3493.47 $\pm$ 2973.51 | 0.55                 |
| DMS            | ESM-2-150M     | 9387.26 $\pm$ 6127.43         | 5098.11 $\pm$ 3667.28 | $6.2 \times 10^{-3}$ |
| DMS            | ESM-2-650M     | 8562.32 $\pm$ 6869.83         | 5143.42 $\pm$ 3701.19 | $1.3 \times 10^{-2}$ |
| DMS            | ESM-2-3B       | 9130.00 $\pm$ 6500.32         | 5037.16 $\pm$ 3586.46 | $3.1 \times 10^{-3}$ |
| Alpha          | Hie-BiLSTM     | 7032.43 $\pm$ 5084.98         | 3567.86 $\pm$ 3281.72 | 0.11                 |
| Alpha          | ESM-2-150M     | 10427.86 $\pm$ 6253.14        | 6732.86 $\pm$ 4672.34 | 0.19                 |
| Alpha          | ESM-2-650M     | 12948.43 $\pm$ 5753.63        | 6816.29 $\pm$ 4391.33 | 0.11                 |
| Alpha          | ESM-2-3B       | 10766.14 $\pm$ 5597.26        | 6480.71 $\pm$ 4743.41 | 0.11                 |
| Beta           | Hie-BiLSTM     | 4046.00 $\pm$ 2348.84         | 1339.75 $\pm$ 1893.99 | $2.7 \times 10^{-2}$ |
| Beta           | ESM-2-150M     | 7902.00 $\pm$ 5731.57         | 3545.25 $\pm$ 3618.94 | $7.8 \times 10^{-3}$ |
| Beta           | ESM-2-650M     | 7932.25 $\pm$ 4493.72         | 3368.38 $\pm$ 3302.71 | $2.0 \times 10^{-2}$ |
| Beta           | ESM-2-3B       | 5013.25 $\pm$ 2767.09         | 2697.88 $\pm$ 2765.35 | $2.7 \times 10^{-2}$ |
| Delta          | Hie-BiLSTM     | 8102.29 $\pm$ 7945.03         | 5107.57 $\pm$ 7242.10 | 0.15                 |
| Delta          | ESM-2-150M     | 11119.86 $\pm$ 5348.50        | 7255.43 $\pm$ 4137.39 | $7.8 \times 10^{-3}$ |
| Delta          | ESM-2-650M     | 11190.29 $\pm$ 5828.41        | 7094.71 $\pm$ 3428.05 | $3.9 \times 10^{-2}$ |
| Delta          | ESM-2-3B       | 10644.57 $\pm$ 4260.09        | 6154.71 $\pm$ 2643.14 | $2.3 \times 10^{-2}$ |
| Omicron        | Hie-BiLSTM     | 7655.30 $\pm$ 5208.90         | 3258.77 $\pm$ 3557.71 | $7.0 \times 10^{-6}$ |
| Omicron        | ESM-2-150M     | 10366.23 $\pm$ 5749.98        | 4261.67 $\pm$ 3283.74 | $7.0 \times 10^{-6}$ |
| Omicron        | ESM-2-650M     | 8687.47 $\pm$ 5740.13         | 4140.77 $\pm$ 3198.91 | $1.5 \times 10^{-5}$ |
| Omicron        | ESM-2-3B       | 8680.50 $\pm$ 6125.22         | 4059.87 $\pm$ 3336.31 | $8.5 \times 10^{-5}$ |
| Combined       | Hie-BiLSTM     | 6259.70 $\pm$ 5377.95         | 3841.07 $\pm$ 4183.94 | $2.4 \times 10^{-4}$ |
| Combined       | ESM-2-150M     | 9815.72 $\pm$ 6052.50         | 5308.55 $\pm$ 4280.29 | $1.3 \times 10^{-6}$ |
| Combined       | ESM-2-650M     | 8857.37 $\pm$ 6263.02         | 5326.83 $\pm$ 4195.95 | $1.6 \times 10^{-5}$ |
| Combined       | ESM-2-3B       | 8735.83 $\pm$ 6094.85         | 5204.23 $\pm$ 4054.37 | $8.6 \times 10^{-6}$ |



**Figure S4: Rank distributions across validation sets.** Box plots compare CSCS, CAC, CLIMB-only ablation, and random baselines.

**Table S6: Representative optimal-weight trajectories.**



**Table S7: Domain-resolved average rank shifts for all possible mutations.** Negative values indicate promotion by CAC relative to CSCS.

| Random                                    |                    |                    |                    |                    |                    |                    |
|-------------------------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
|                                           | NTD                | RBD                | FP                 | HR1                | HR2                | TM                 |
| Rank <sub>CSCS</sub>                      | 11944.52 ± 7004.79 | 12133.53 ± 6969.43 | 12123.73 ± 7257.53 | 12087.71 ± 6901.65 | 12175.62 ± 7103.16 | 11885.03 ± 6941.11 |
| Rank <sub>CAC</sub>                       | 12171.62 ± 7025.52 | 12080.15 ± 6974.83 | 12630.38 ± 7010.15 | 12142.85 ± 6900.83 | 11685.95 ± 6777.75 | 12505.88 ± 7120.62 |
| Rank <sub>CAC</sub> -Rank <sub>CSCS</sub> | 227.10 ± 132.97    | -53.37 ± 151.48    | 506.65 ± 531.07    | 55.14 ± 262.06     | -489.66 ± 315.40   | 620.85 ± 465.67    |

| Hie                                       |                    |                    |                    |                    |                    |                    |
|-------------------------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
|                                           | NTD                | RBD                | FP                 | HR1                | HR2                | TM                 |
| Rank <sub>CSCS</sub>                      | 10466.20 ± 6778.14 | 11202.43 ± 6765.50 | 10149.65 ± 6614.45 | 14189.60 ± 6694.64 | 15227.74 ± 6127.75 | 16996.08 ± 6290.81 |
| Rank <sub>CAC</sub>                       | 11348.84 ± 6873.78 | 11390.06 ± 6772.49 | 12108.50 ± 6787.24 | 13337.70 ± 7197.38 | 13155.33 ± 6870.01 | 15140.41 ± 7737.44 |
| Rank <sub>CAC</sub> -Rank <sub>CSCS</sub> | 882.64 ± 129.38    | 187.63 ± 147.07    | 1958.85 ± 498.80   | -851.90 ± 263.93   | -2072.41 ± 295.73  | -1855.68 ± 466.99  |

| ESM2 150M                                 |                    |                    |                    |                    |                    |                    |
|-------------------------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
|                                           | NTD                | RBD                | FP                 | HR1                | HR2                | TM                 |
| Rank <sub>CSCS</sub>                      | 13023.94 ± 6808.17 | 11607.07 ± 6531.08 | 12980.91 ± 7456.75 | 13490.32 ± 8041.74 | 9438.89 ± 6976.61  | 8834.14 ± 5628.47  |
| Rank <sub>CAC</sub>                       | 12572.51 ± 7145.65 | 11886.74 ± 6891.00 | 12471.28 ± 6987.12 | 12388.43 ± 6968.37 | 11548.68 ± 6733.08 | 12837.30 ± 7388.89 |
| Rank <sub>CAC</sub> -Rank <sub>CSCS</sub> | -451.43 ± 132.28   | 279.67 ± 145.86    | -509.63 ± 537.83   | -1101.90 ± 285.72  | 2109.79 ± 311.47   | 4003.16 ± 434.97   |

| ESM2 650M                                 |                    |                    |                    |                    |                    |                    |
|-------------------------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
|                                           | NTD                | RBD                | FP                 | HR1                | HR2                | TM                 |
| Rank <sub>CSCS</sub>                      | 12811.63 ± 6538.17 | 9723.82 ± 6043.81  | 13007.99 ± 7165.56 | 15291.89 ± 7661.06 | 11530.56 ± 7362.05 | 8875.89 ± 5540.36  |
| Rank <sub>CAC</sub>                       | 11855.36 ± 6953.31 | 11731.33 ± 6895.52 | 12743.82 ± 7033.22 | 13017.13 ± 7077.25 | 12031.24 ± 6870.96 | 13006.45 ± 7390.47 |
| Rank <sub>CAC</sub> -Rank <sub>CSCS</sub> | -956.27 ± 127.92   | 2007.51 ± 140.87   | -264.17 ± 528.45   | -2274.75 ± 280.05  | 500.67 ± 323.50    | 4130.56 ± 432.54   |

| ESM2 3B                                   |                    |                    |                    |                    |                    |                    |
|-------------------------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
|                                           | NTD                | RBD                | FP                 | HR1                | HR2                | TM                 |
| Rank <sub>CSCS</sub>                      | 10815.43 ± 5820.03 | 8636.73 ± 5726.55  | 13979.20 ± 7093.41 | 17916.36 ± 6658.11 | 12504.48 ± 7011.04 | 11767.12 ± 7210.61 |
| Rank <sub>CAC</sub>                       | 11340.45 ± 6759.74 | 11234.81 ± 6714.19 | 13450.11 ± 7191.18 | 13688.97 ± 7163.69 | 12675.30 ± 7068.91 | 13522.60 ± 7594.00 |
| Rank <sub>CAC</sub> -Rank <sub>CSCS</sub> | 525.02 ± 119.55    | 2598.08 ± 135.57   | -529.09 ± 531.63   | -4227.39 ± 262.60  | 170.82 ± 319.84    | 1755.48 ± 490.39   |

**Table S8: Domain classification of mutations in the combined validation set.**

| Validation sets of escape mutations per domains |                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NTD                                             | (1) L18F, (2) T19R, (3) A67V, (4) D80A, (5) T95I, (6) G142D, (7) L212I, (8) D215G                                                                                                                                                                                                                                                             |
| RBD                                             | (9) G339D, (10) S371L, (11) S373P, (12) S375F, (13) K417E, (14) K417N, (15) N440K, (16) N440Q, (17) V445A, (18) G446S, (19) N450D, (20) L452R, (21) Y453F, (22) L455F, (23) S477N, (24) T478K, (25) E484A, (26) E484K, (27) G485D, (28) F486V, (29) F490L, (30) F490S, (31) Q493K, (32) Q493R, (33) G496S, (34) Q498R, (35) N501Y, (36) Y505H |
| FP                                              | (52) D796Y                                                                                                                                                                                                                                                                                                                                    |
| HR1                                             | (54) D950N, (55) Q954H, (56) N969K, (57) L981F, (58) S982A                                                                                                                                                                                                                                                                                    |
| HR2                                             |                                                                                                                                                                                                                                                                                                                                               |
| TM                                              |                                                                                                                                                                                                                                                                                                                                               |
| Others (Not in domains)                         | (37) T547K, (38) A670D, (39) D614G, (40) H655Y, (41) N679K, (42) P681H, (43) P681R, (44) R682Q, (45) R685S, (46) V687G, (47) A701V, (48) T716I, (49) N764K, (50) G769E, (51) Q779K, (59) D1118H, (60) V1128A                                                                                                                                  |

**Table S9: Domain-specific rank changes for the combined validation set.** This table focuses only on the 60 known substitutions in the combined set.

| Random                                        |                    |                    |                |                    |      |      |
|-----------------------------------------------|--------------------|--------------------|----------------|--------------------|------|------|
|                                               | NTD                | RBD                | FP             | HR1                | HR2  | TM   |
| <b>Rank<sub>CSCS</sub></b>                    | 9871.75 ± 7505.33  | 12167.21 ± 6381.07 | 994.00 ± 0.00  | 14502.40 ± 6923.04 | None | None |
| <b>Rank<sub>CAC</sub></b>                     | 5000.38 ± 4667.58  | 4951.39 ± 3882.43  | 570.00 ± 0.00  | 6596.40 ± 6387.73  | None | None |
| <b>Rank<sub>CAC</sub>-Rank<sub>CSCS</sub></b> | -4871.38 ± 3735.63 | -7215.82 ± 1411.48 | -424.00 ± 0.00 | -7906.00 ± 3823.49 | None | None |

| Hie                                           |                    |                    |                |                    |      |      |
|-----------------------------------------------|--------------------|--------------------|----------------|--------------------|------|------|
|                                               | NTD                | RBD                | FP             | HR1                | HR2  | TM   |
| <b>Rank<sub>CSCS</sub></b>                    | 6750.88 ± 7516.32  | 6165.96 ± 4886.01  | 1312.00 ± 0.00 | 9186.20 ± 7027.47  | None | None |
| <b>Rank<sub>CAC</sub></b>                     | 4375.88 ± 7437.02  | 3971.32 ± 3881.44  | 393.00 ± 0.00  | 6076.40 ± 5016.69  | None | None |
| <b>Rank<sub>CAC</sub>-Rank<sub>CSCS</sub></b> | -2375.00 ± 3133.41 | -2194.64 ± 1158.17 | -919.00 ± 0.00 | -3109.80 ± 4108.29 | None | None |

| ESM2 150M                                     |                    |                    |                  |                    |      |      |
|-----------------------------------------------|--------------------|--------------------|------------------|--------------------|------|------|
|                                               | NTD                | RBD                | FP               | HR1                | HR2  | TM   |
| <b>Rank<sub>CSCS</sub></b>                    | 11047.12 ± 7122.90 | 7464.82 ± 4899.32  | 17044.00 ± 0.00  | 10160.40 ± 8054.21 | None | None |
| <b>Rank<sub>CAC</sub></b>                     | 4911.50 ± 4695.86  | 4774.75 ± 3699.31  | 585.00 ± 0.00    | 6564.40 ± 6111.78  | None | None |
| <b>Rank<sub>CAC</sub>-Rank<sub>CSCS</sub></b> | -6135.62 ± 2980.30 | -2690.07 ± 1163.19 | -16459.00 ± 0.00 | -3653.60 ± 4521.60 | None | None |

| ESM2 650M                                     |                    |                    |                  |                    |      |      |
|-----------------------------------------------|--------------------|--------------------|------------------|--------------------|------|------|
|                                               | NTD                | RBD                | FP               | HR1                | HR2  | TM   |
| <b>Rank<sub>CSCS</sub></b>                    | 8481.62 ± 6743.16  | 6187.43 ± 4179.92  | 21296.00 ± 0.00  | 9982.80 ± 8609.31  | None | None |
| <b>Rank<sub>CAC</sub></b>                     | 4773.88 ± 4507.99  | 4789.71 ± 3725.69  | 644.00 ± 0.00    | 6564.40 ± 6111.78  | None | None |
| <b>Rank<sub>CAC</sub>-Rank<sub>CSCS</sub></b> | -3707.75 ± 2815.26 | -1397.71 ± 1032.81 | -20652.00 ± 0.00 | -3418.40 ± 4667.96 | None | None |

| ESM2 3B                                       |                    |                   |                  |                    |      |      |
|-----------------------------------------------|--------------------|-------------------|------------------|--------------------|------|------|
|                                               | NTD                | RBD               | FP               | HR1                | HR2  | TM   |
| <b>Rank<sub>CSCS</sub></b>                    | 7524.50 ± 6152.67  | 4987.25 ± 3417.16 | 19945.00 ± 0.00  | 12459.60 ± 7111.16 | None | None |
| <b>Rank<sub>CAC</sub></b>                     | 4453.62 ± 4234.99  | 4567.21 ± 3520.72 | 1901.00 ± 0.00   | 6652.80 ± 5901.62  | None | None |
| <b>Rank<sub>CAC</sub>-Rank<sub>CSCS</sub></b> | -3070.88 ± 2544.85 | -420.04 ± 698.70  | -18044.00 ± 0.00 | -5806.80 ± 3610.76 | None | None |

```
$ conda create --name climb python=3.10 -y
$ conda activate climb
$ pip install -r requirements.txt
```

2. Generate primary scores.

```
$ cd viral-analyzer
$ python main.py \
  --escape "SARS-CoV-2-WildType>*" \
  --result "SARS-CoV-2-WildType-*" \
  --scores --seed 42
```

3. Optimize component weights.

```
$ python main.py \
  --escape "SARS-CoV-2-WildType>*" \
  --result "SARS-CoV-2-WildType-*" \
  --minimizer --seed 42
```

4. Run the CLIMB-only ablation.

```
$ python main.py \
  --escape "SARS-CoV-2-WildType>*" \
  --result "SARS-CoV-2-WildType-Random" \
  --ablation --seed 42
```

## References

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- [2] Brian Hie et al. “Learning the language of viral evolution and escape”. In: *Science* 371.6526 (2021), pp. 284–288.
- [3] Kijong Yi et al. “Mutational spectrum of SARS-CoV-2 during the global pandemic”. In: *Experimental & Molecular Medicine* 53.8 (Aug. 2021), pp. 1229–1237. ISSN: 1226-3613. DOI: 10.1038/s12276-021-00658-z. URL: <http://dx.doi.org/10.1038/s12276-021-00658-z>.
- [4] Zeming Lin et al. “Evolutionary-scale prediction of atomic-level protein structure with a language model”. In: *Science* 379.6637 (2023), pp. 1123–1130.